

MBA TESTS DATA INSIGHTS REPORT

1. Introduction

Objective

The purpose of this analysis is to evaluate candidate performance in MBA admission tests across multiple aptitude sections.

The goals are:

- To understand performance distribution and identify outliers.
- To explore relationships among different aptitude sections.
- To examine behavioral and contextual factors such as test duration, tab switching, and proctoring.
- To generate actionable insights for test design, candidate preparation, and program improvement.

Dataset Scope

The dataset consists of candidate-level test submissions with:

- Demographics: (e.g., name, email, gender, phone) → excluded for privacy.
- System Metadata: browser, OS, proctoring flags.
- Test Metadata: start and submit times, calculated duration, test status, tab switching.
- Scores: overall test score and section-wise scores

Key Challenges

- Personally Identifiable Information.
- Mixed data formats (e.g., percentages stored as strings).
- Possible duplicates due to multiple test attempts.
- Outliers in test duration and scores.
- Proctoring data with unclear meaning.

2. Data Cleaning and Preparation

- PII Handling: For analysis, PII fields (name, phone, email, etc.) were not used in visualizations or feature engineering. They were retained in the dataset only for internal reference.”
- Standardization: Converted percentages to numeric; verified valid ranges; parsed timestamps into durations.
- Duplicates: Retained only the latest submission per candidate (Completed status preferred).
- Missing Values: Kept missing section scores but excluded them from averages.
- Outliers: Flagged extreme values but only removed invalid ones (e.g., >100%).
- Final Dataset: Only Completed test records, with cohort identifiers, metadata, and all section scores.

3. Feature Engineering

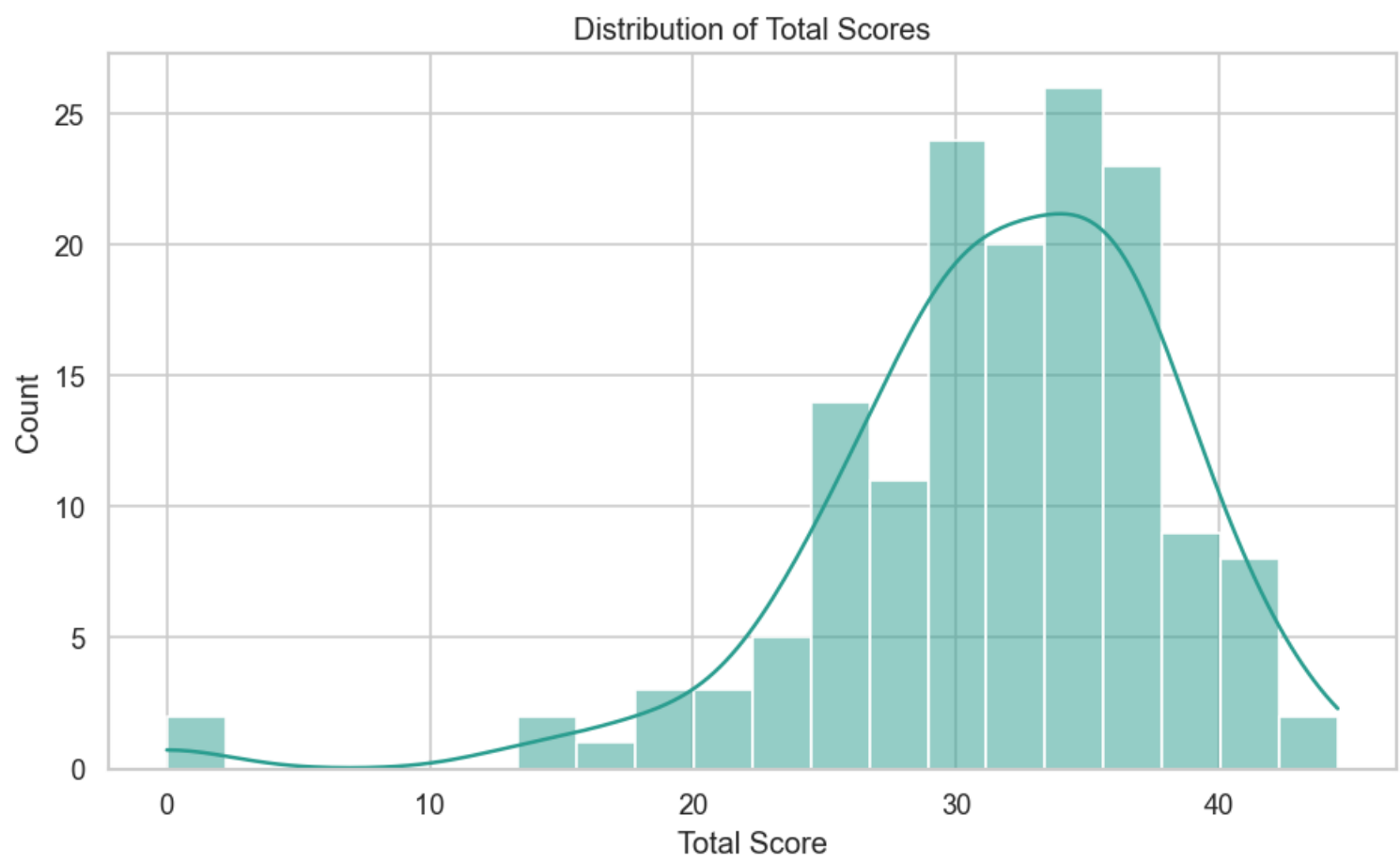
New derived fields:

- total_percentage_num: numeric overall percentage.
- section_percentage_num: numeric version for each section.
- score_ratio: candidate_proct ÷ section_max_score.
- duration_minutes: cleaned test duration in minutes.
- tabswitch_flag: 1 if tab switch > 0, else 0.
- proctor_flag: 1 if proctoring analysis above median. (proctoring_analysis > median; proctoring_risk is a categorical version (Low/Medium/Higher)) From proctoring_analysis (numeric values), we derived two fields: proctor_flag (binary, 1 if above median) and proctoring_risk (categorical: Low/Medium/Higher)
- cohort_batch: parsed college and batch separately.

4. Exploratory Data Analysis (EDA)

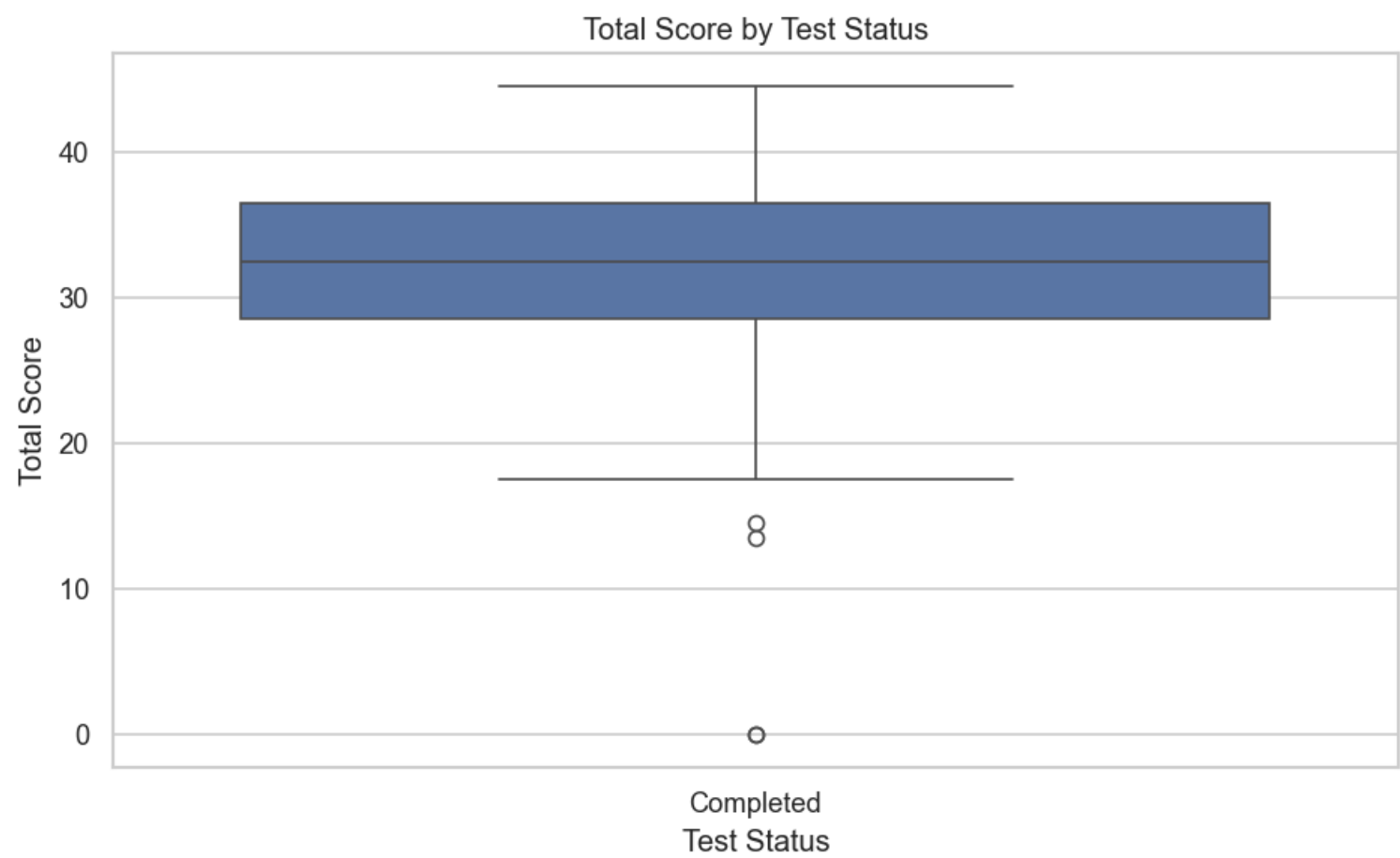
4.1 Overall Performance

- Figure 1: Distribution of Total Scores



Shows candidate overall performance spread, with most clustered between 50–75%.

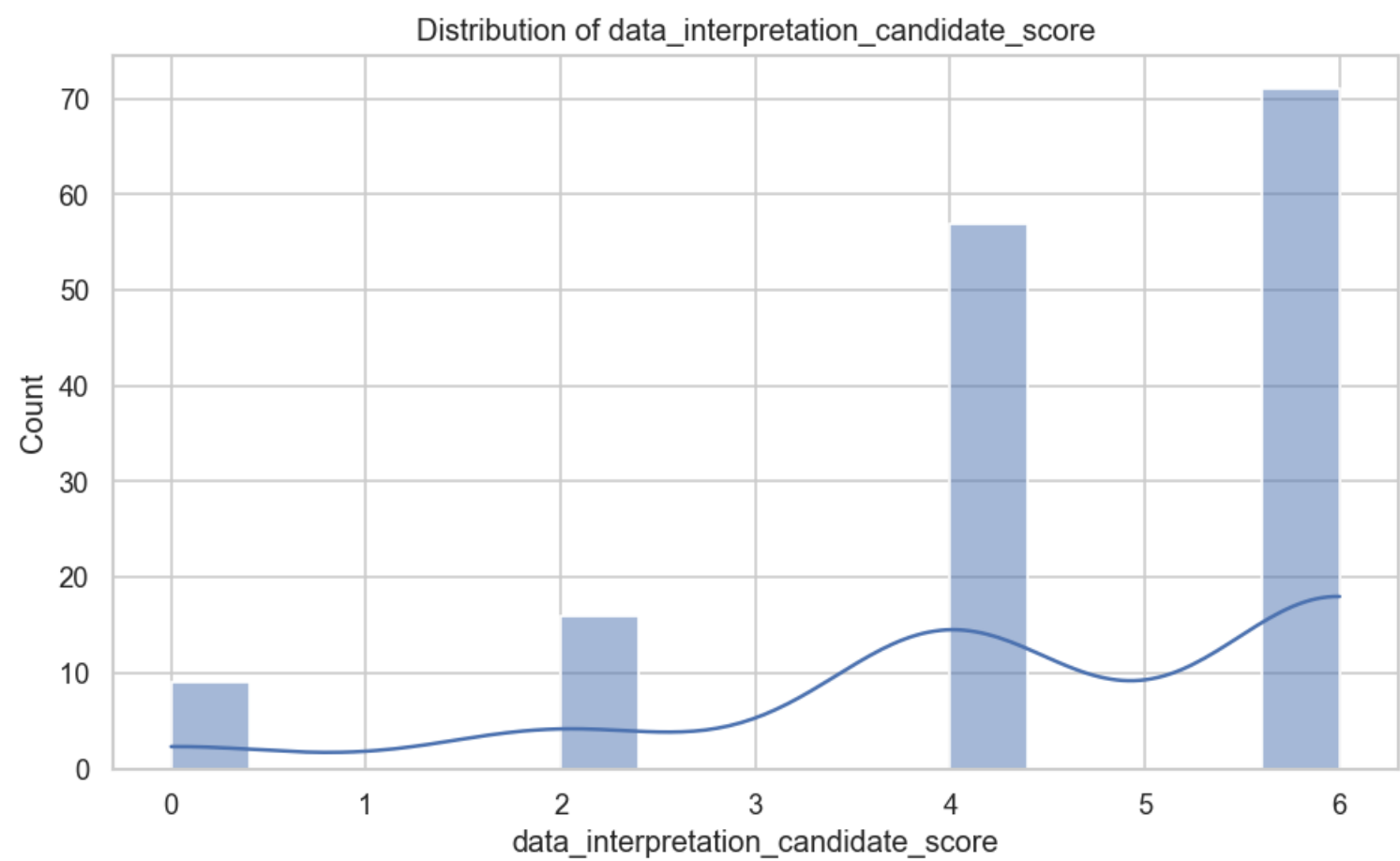
- Figure 2: Boxplot of Test Status vs Scores



Confirms consistency among candidates who completed tests.

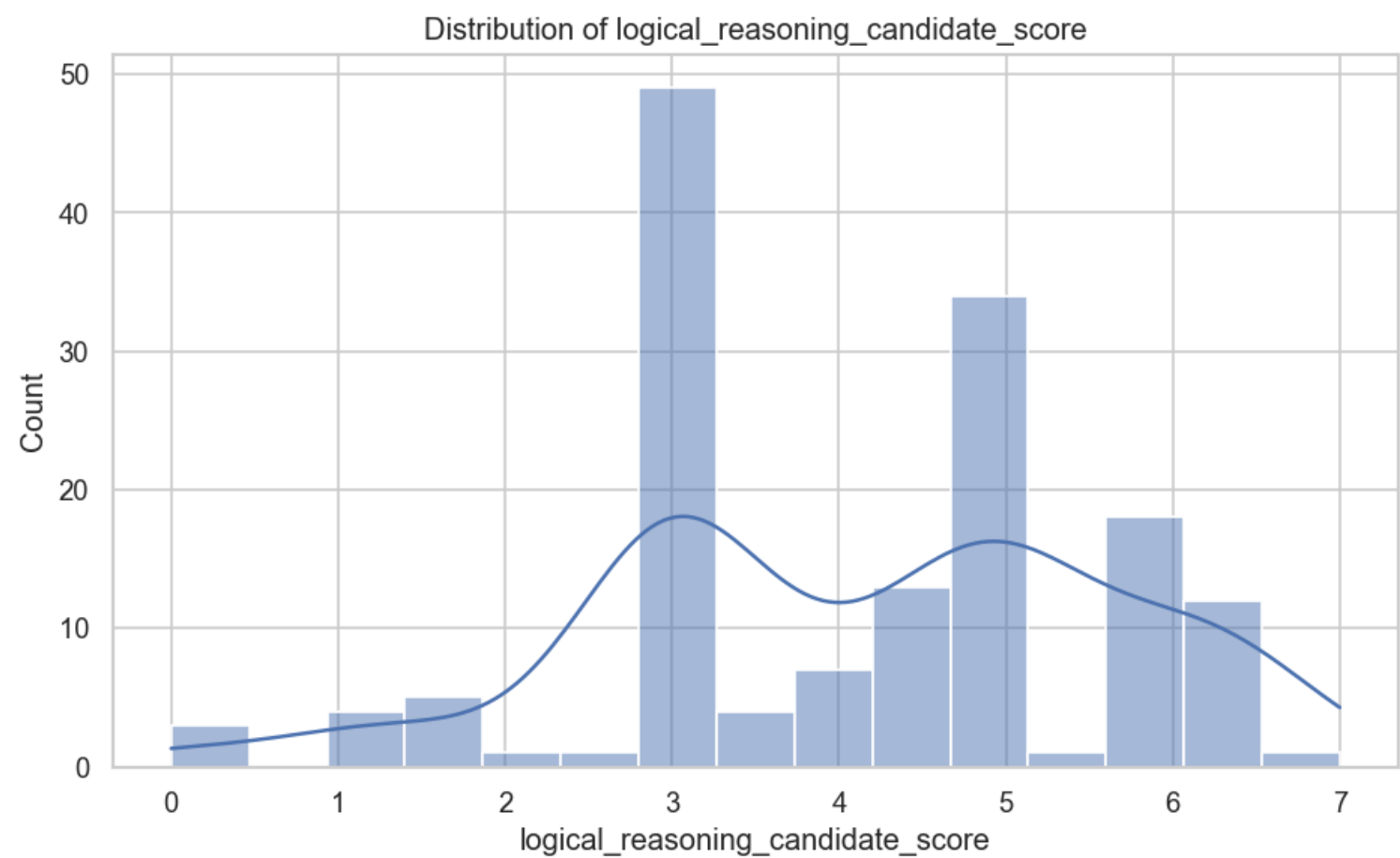
4.2 Section-wise Performance

- Figure 3: Data Interpretation Score Distribution



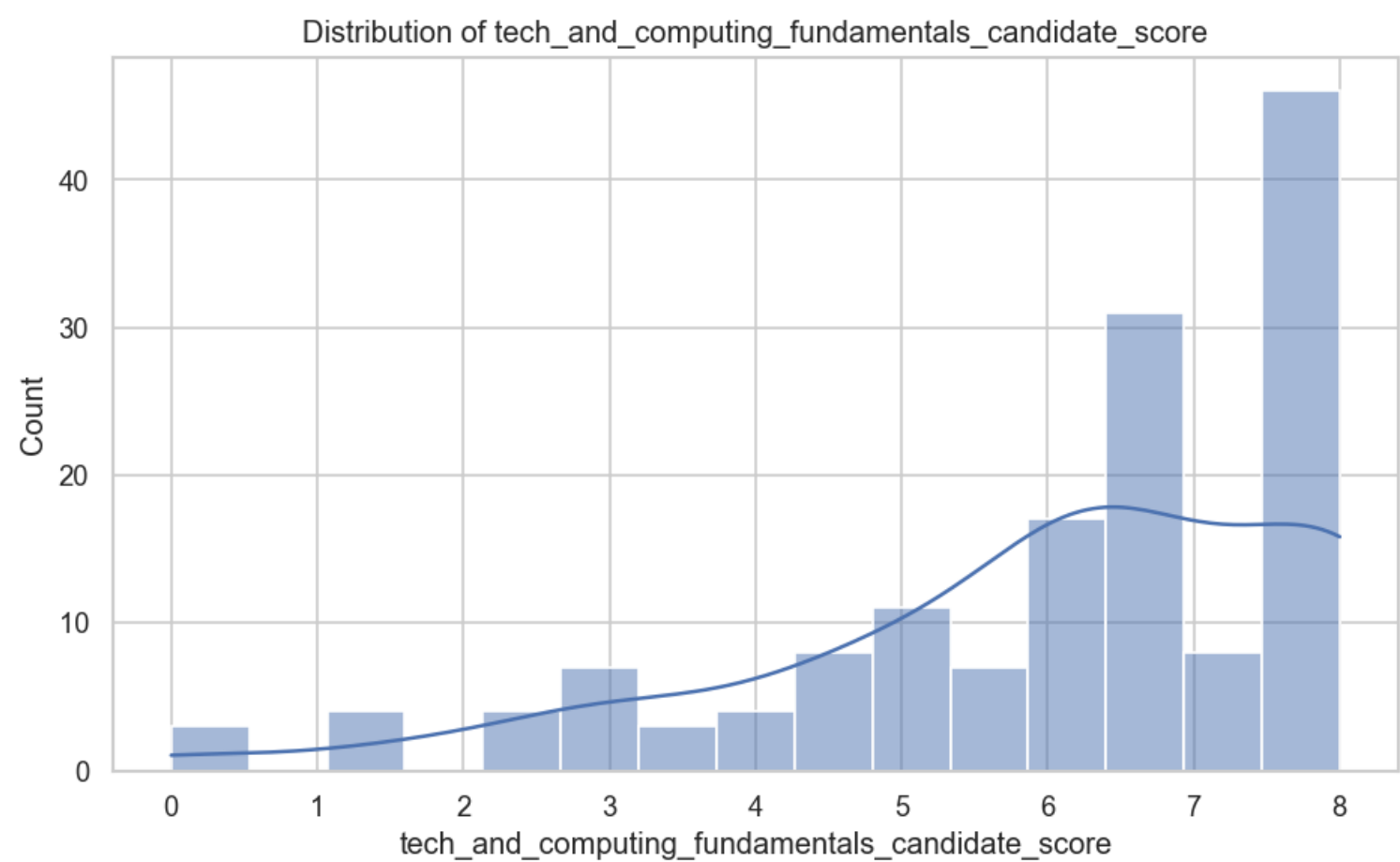
Stronger section overall, many scoring above 60%.

• Figure 4: Logical Reasoning Score Distribution



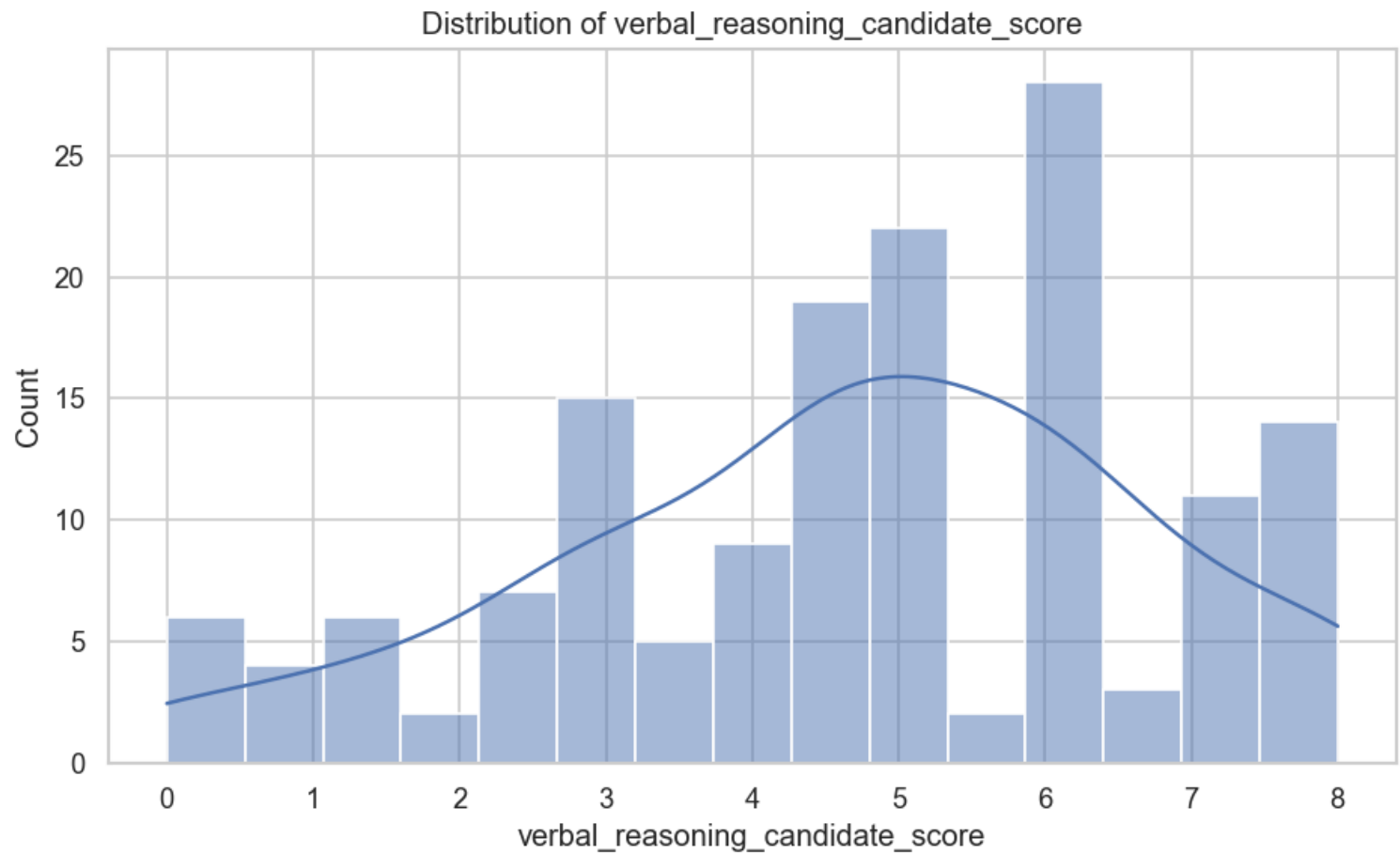
Balanced performance, clustered around mid-60s.

• Figure 5: Tech & Computing Fundamentals Score Distribution



Bimodal results, indicating different background levels.

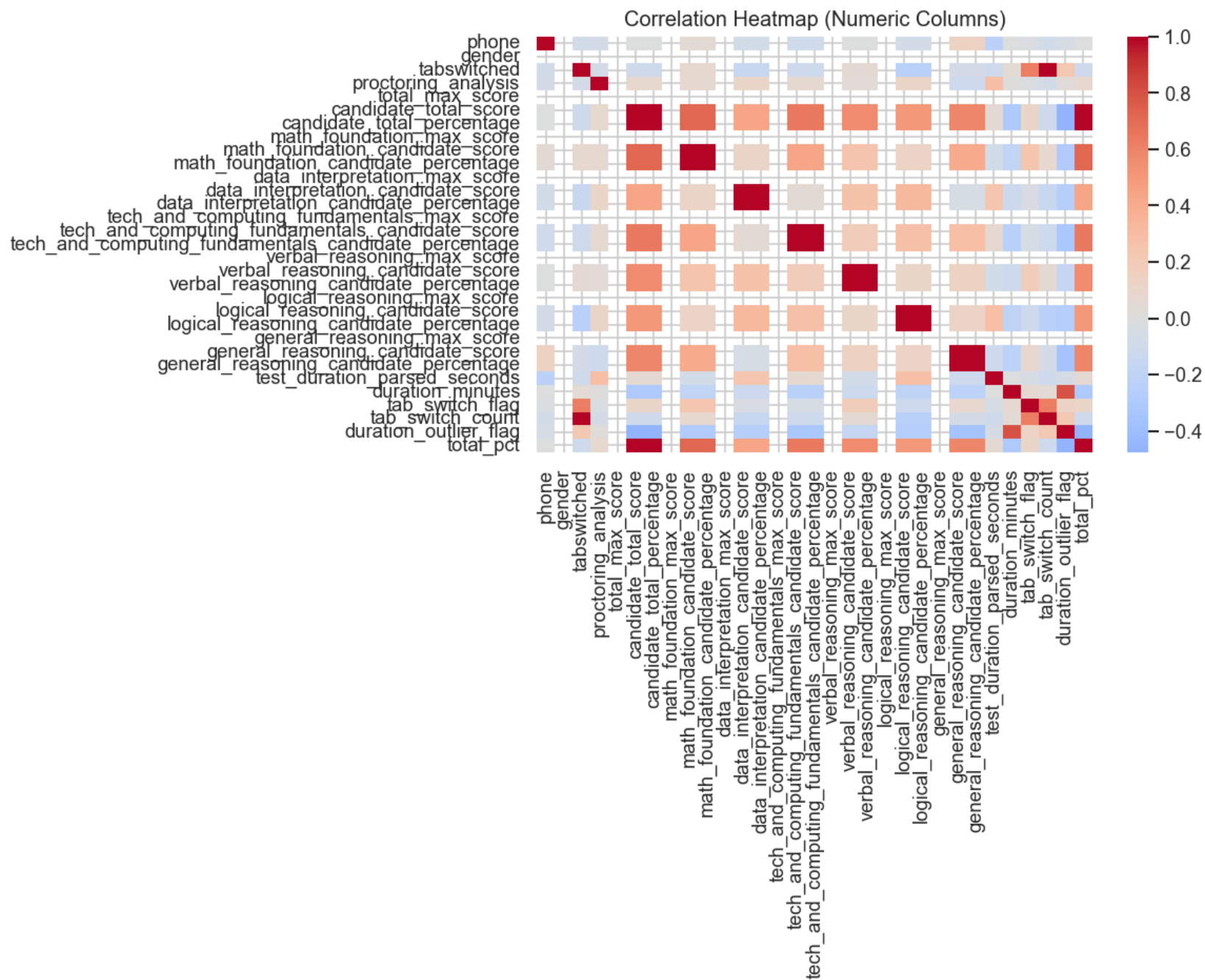
• Figure 6: Verbal Reasoning Score Distribution



Higher scores concentrated toward 80–100%.

4.3 Section Comparisons & Correlations

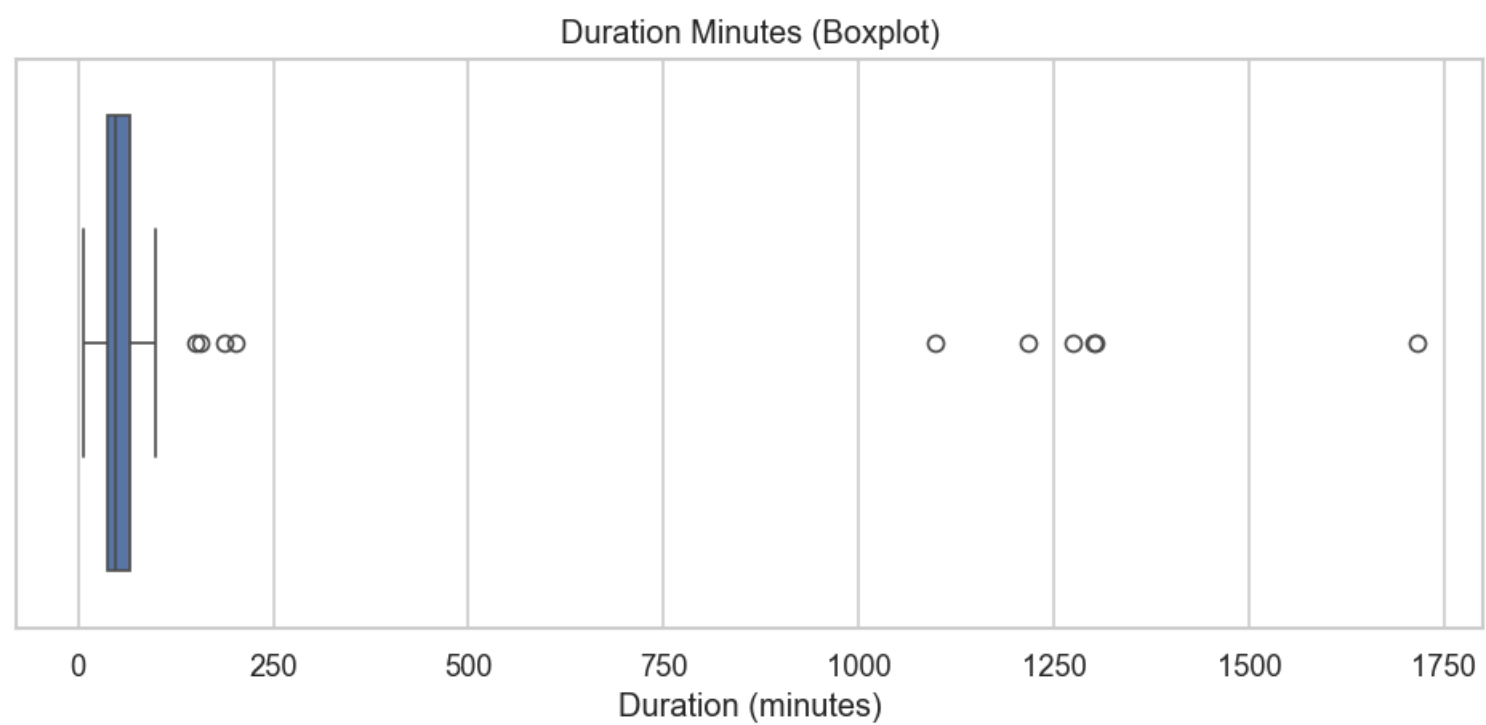
• Figure 7: Correlation Heatmap of Section Scores



Shows Verbal, Logical, and General Reasoning strongly correlated with overall results. Math and Tech moderately correlated.

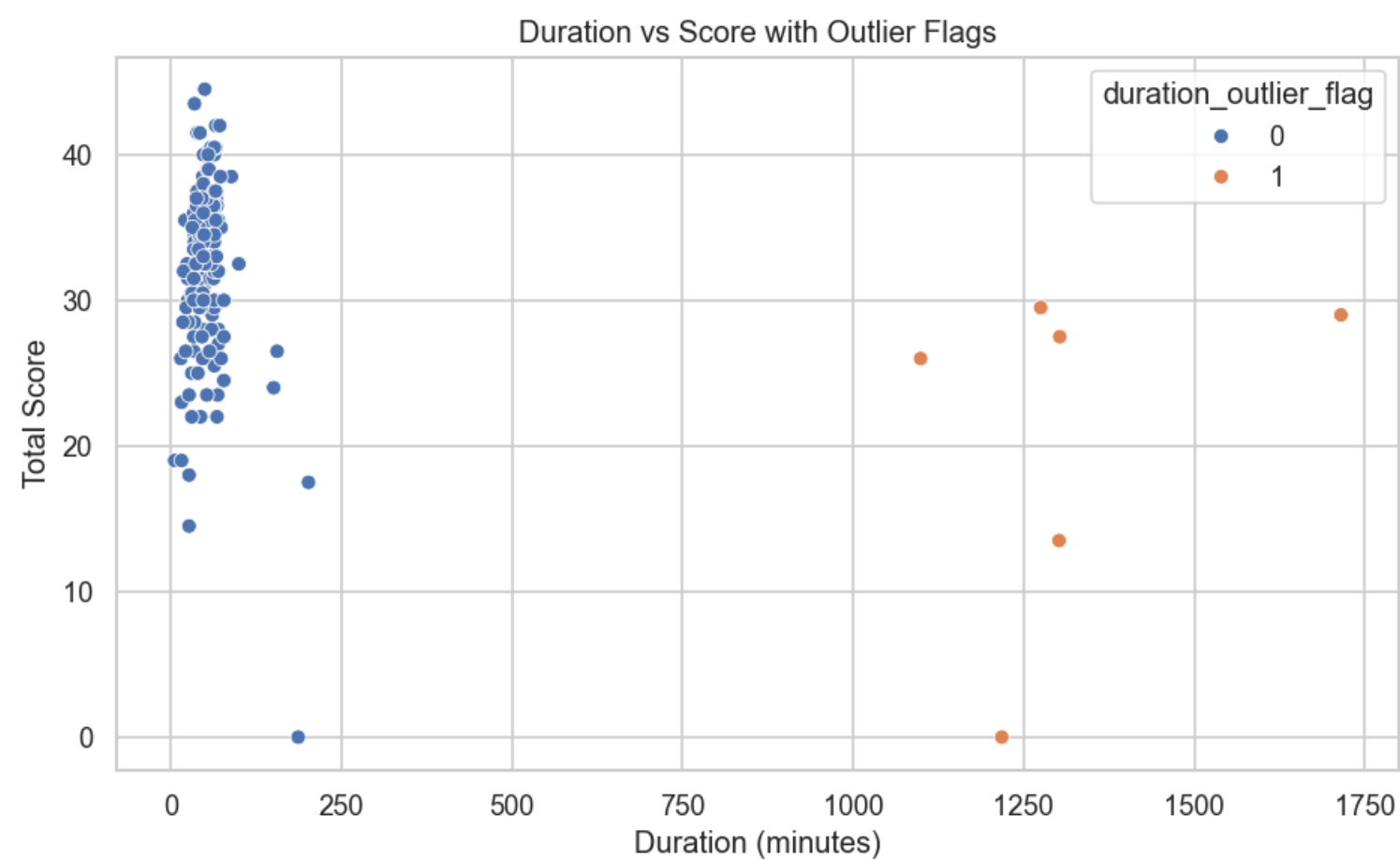
4.4 Time & Behavior Signals

- Figure 8: Distribution of Duration (Boxplot)



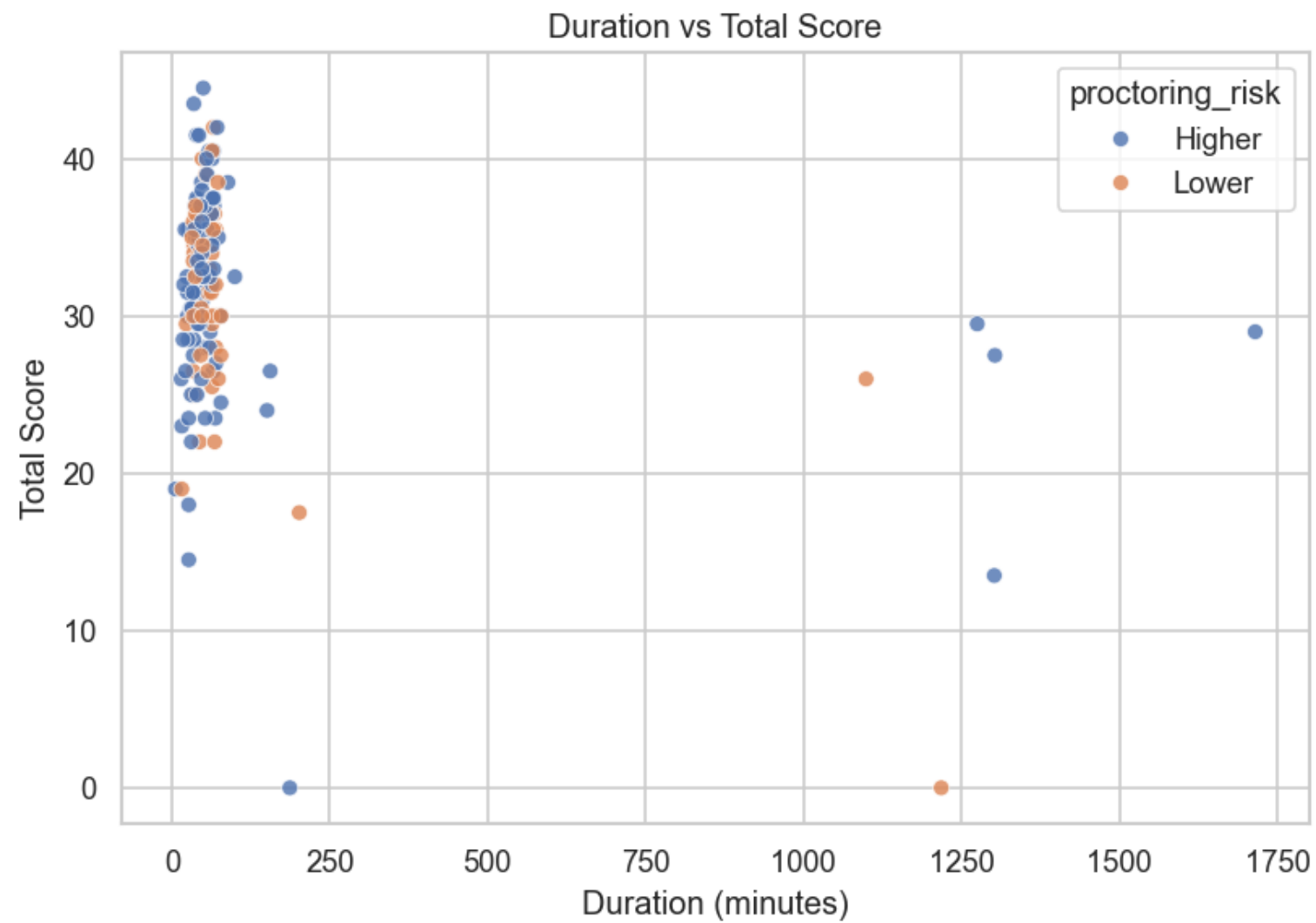
Most candidates completed tests in 45–55 minutes.

- Figure 9: Duration vs Score Scatter (Outlier Check)



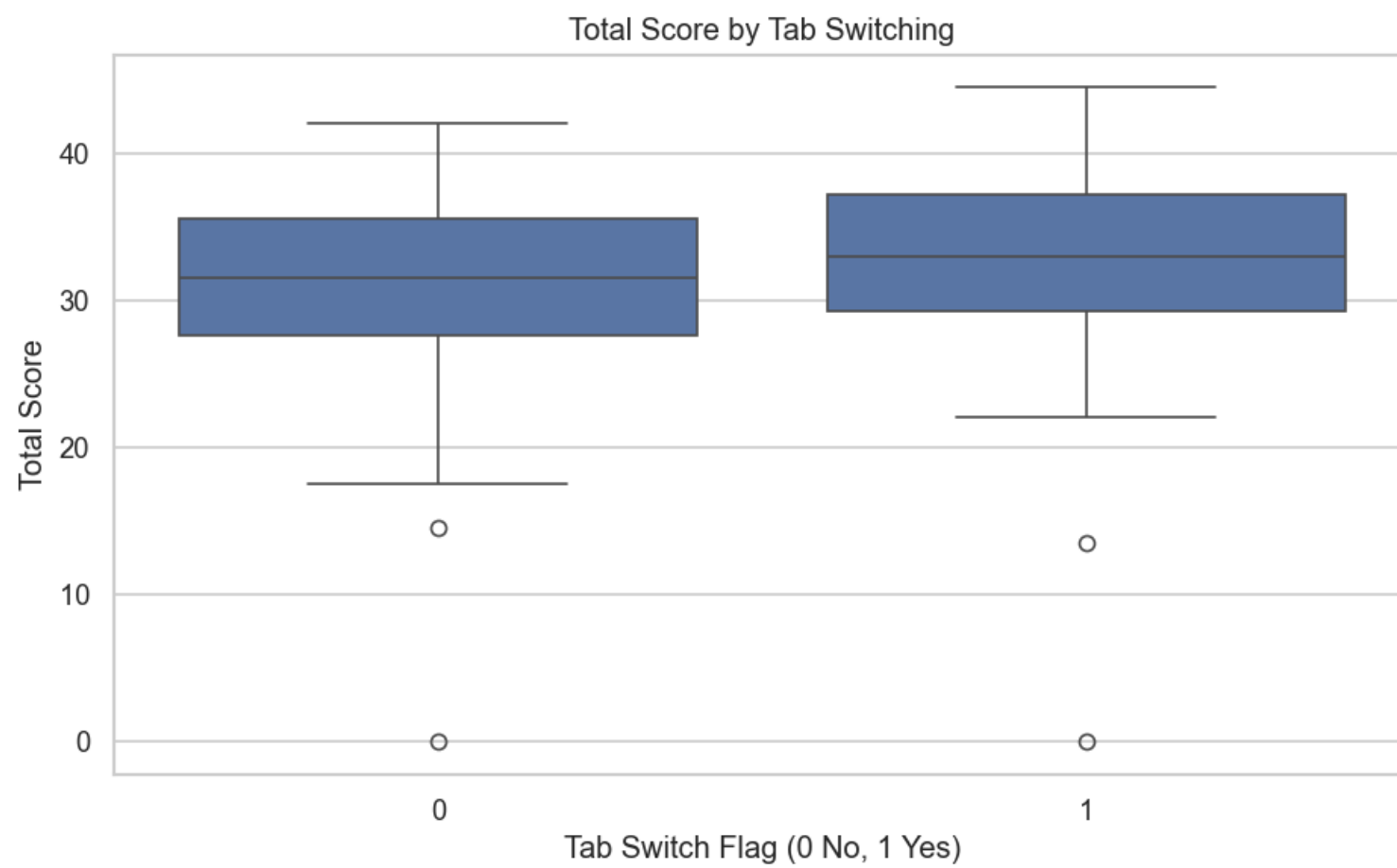
Identifies unusually short or long submissions.

- Figure 10: Duration vs Score Scatter (Proctoring Context)



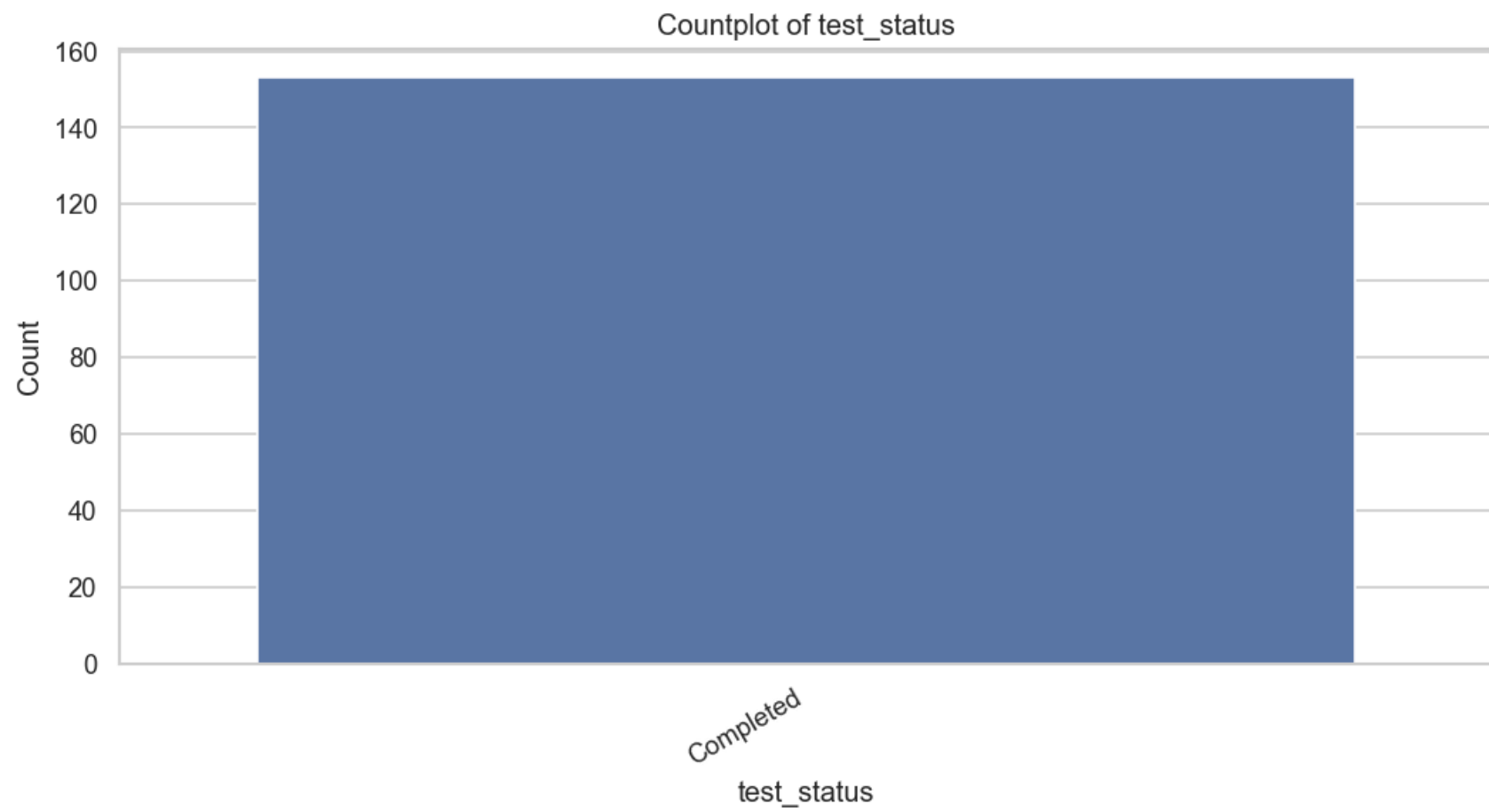
Confirms that extreme durations don't always improve results.

- Figure 11: Boxplot of Tab Switching vs Scores



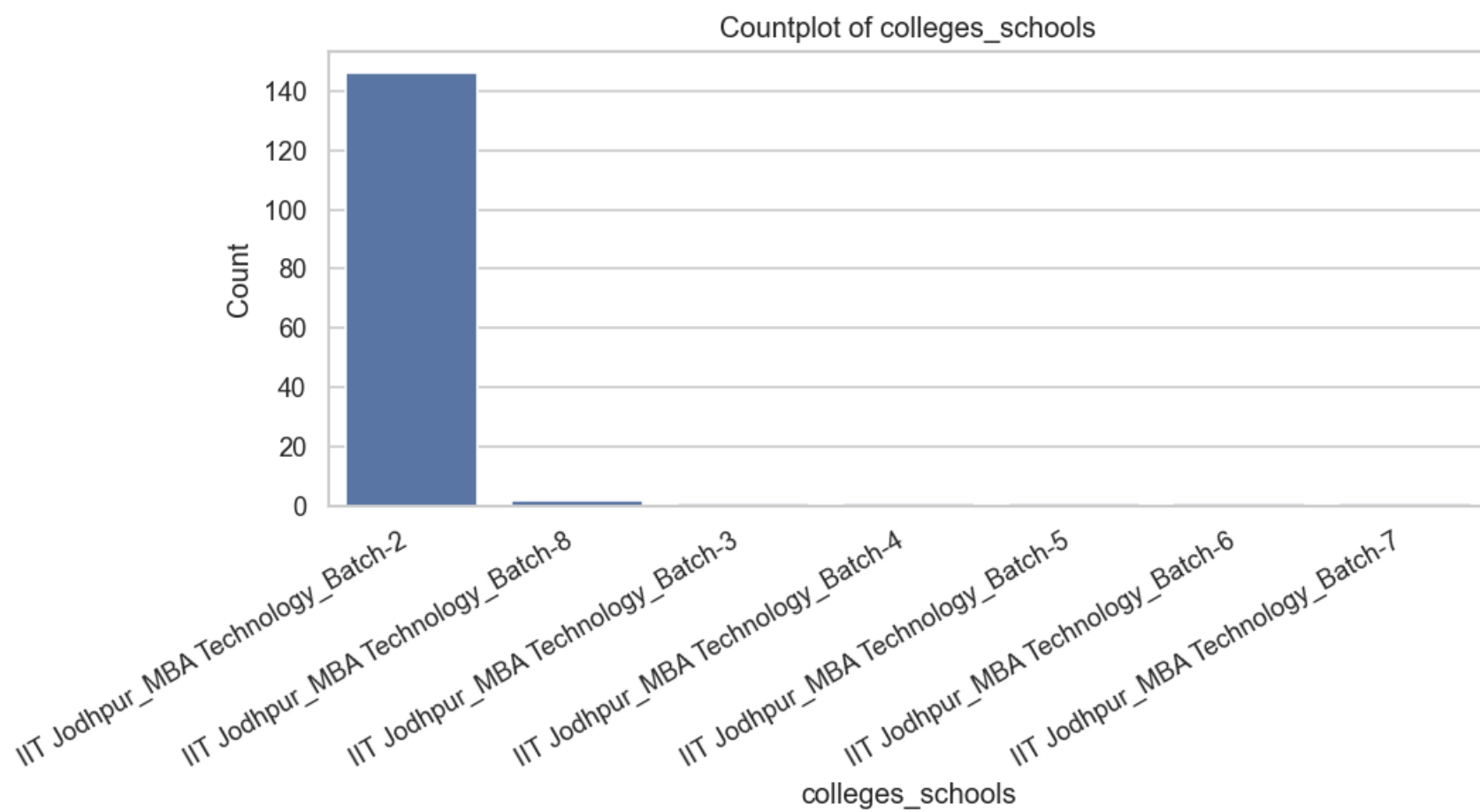
Tab switchers scored slightly lower on average.

- Figure 12: Countplot of Test Status



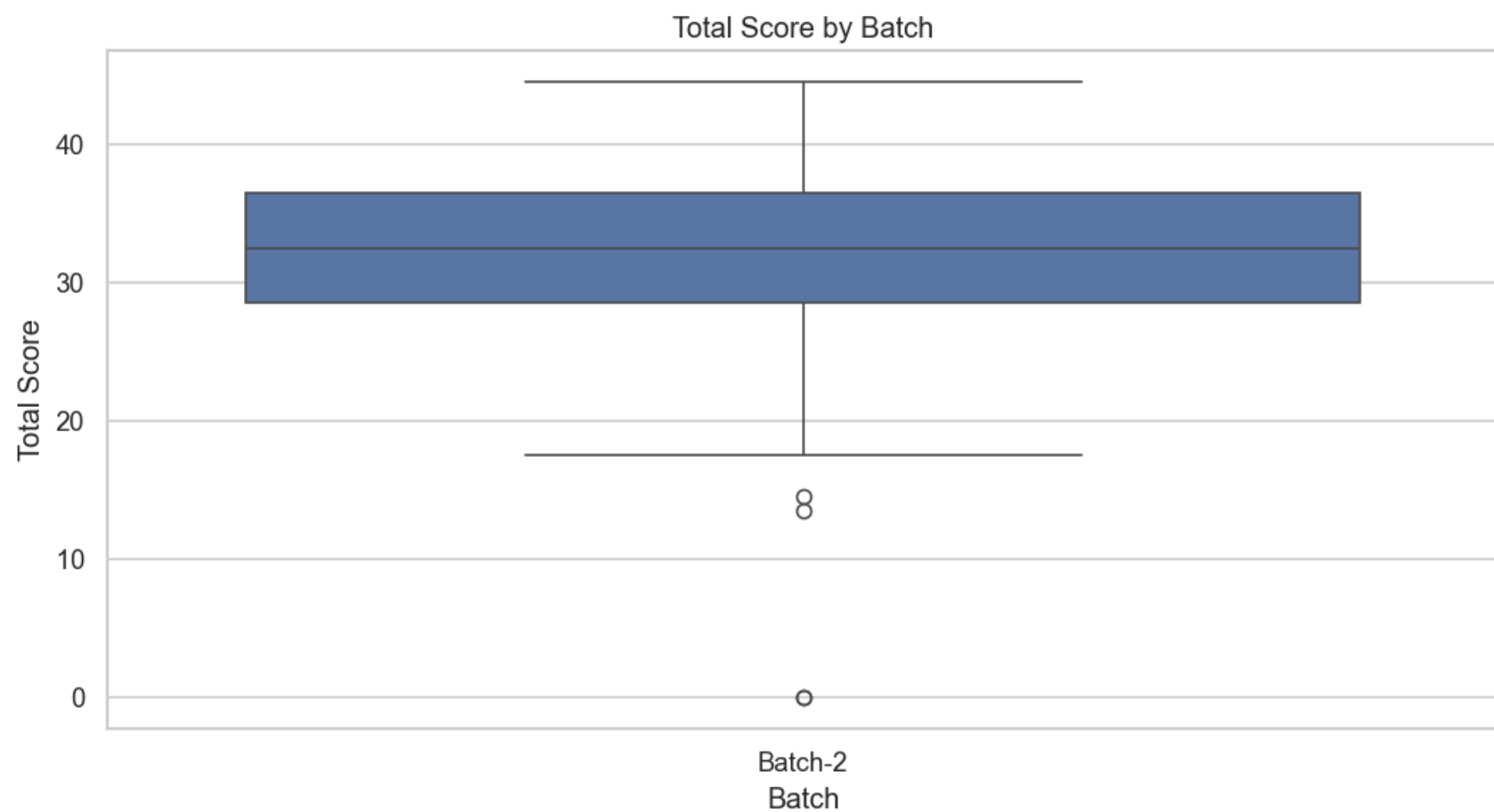
Majority of candidates successfully completed their tests.

- Figure 13: Countplot of Colleges/Schools



Shows representation across different institutions.

- Figure 14: Boxplot of Scores by Batch



Compares scores between batches, differences are minor.

5. Key Insights & Recommendations

1. Performance Levels: Median ~60–65%, majority in 50–75% range.
2. Strengths & Weaknesses:
 - Strongest: Data Interpretation, Verbal Reasoning.
 - Weakest: Math Foundation.
 - Polarized: Tech & Computing.
3. Correlations: Verbal–Logical–General sections strongly tied to overall results.
4. Behavioral Signals: Tab switching and proctoring have weak but noticeable links to lower scores.
5. Recommendations:
 - Strengthen Math Foundation through targeted resources.
 - Offer bridging modules for Tech fundamentals.
 - Challenge high scorers in Verbal & DI to avoid ceiling effects.
 - Align test durations with typical completion ranges.
 - Treat behavioral data (tabs/proctoring) as supportive evidence, not strict penalties.

6. Conclusion

The MBA test dataset highlights overall solid candidate performance, with strengths in Verbal and Data Interpretation and improvement opportunities in Math and Tech. Cohort-level consistency is evident across institutions and batches, and behavioral metadata provides additional context for monitoring. These findings provide practical guidance for improving test design, curriculum support, and candidate preparation.